talenciaglobal

Topological Order

Directed Graphs, Topological Sort (Kahn) · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★★ Advanced

Problem

Produce a topological order of a directed graph - every vertex after all of its prerequisites - choosing the lexicographically smallest such order. If the graph has a cycle, report CYCLE.

Kahn's algorithm with a min-heap repeatedly takes the smallest vertex whose prerequisites are all placed.

What to Compute

- Order: the lexicographically smallest topological order, or CYCLE if none exists.
- Acyclic: Yes if the graph is a DAG, otherwise No.

Input / Output Format

Input: Line 1: N M. Next M lines: a directed edge u v.

Output: Two lines: Order and Acyclic.

Examples

Example 1

Input: 6 6 5 2 5 0 4 0 4 1 2 3 3 1

Output:

```
Order: 4 5 0 2 3 1
Acyclic: Yes
```

Explanation: Both 4 and 5 start with in-degree 0; the min-heap takes 4 first, then 5, and continues to the smallest available vertex at each step.

Example 2

Input: 3 3 0 1 1 2 2 0

Output:

```
Order: CYCLE
Acyclic: No
```

Explanation: The edges form a cycle 0 to 1 to 2 to 0, so no vertex ever reaches in-degree 0 after the start and no topological order exists.

Constraints

- A vertex may appear only after all its prerequisites.
- Break ties by the smallest available vertex (use a min-heap).
- If fewer than N vertices are emitted, report CYCLE.

Sample Run

Sample Input

```
6 6
5 2
5 0
4 0
4 1
2 3
3 1
```

Sample Output

```
Order: 4 5 0 2 3 1
Acyclic: Yes
```